

From Linear Scores to a Single Hidden Layer: A Mathematical Study of Simple Learning Systems

Math4AI Final Capstone

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The Central Question

Our Objective

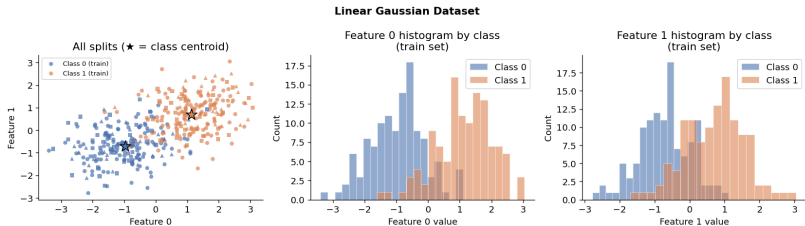
When does a one-hidden-layer nonlinear classifier genuinely improve on a linear decision rule, and when is additional model complexity unnecessary?

Models Compared:

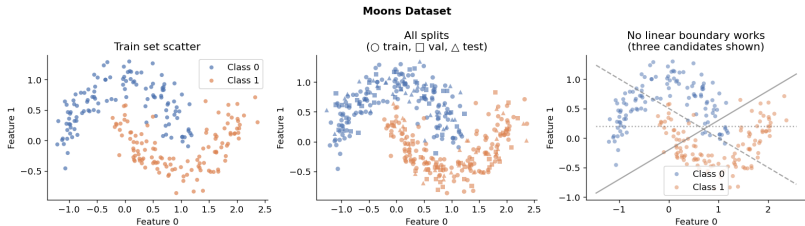
- **Softmax Regression (SR):** Linear score function, affine decision boundaries.
- **1-Hidden-Layer Neural Network (NN):** Non-linear representation via tanh activation, followed by linear classification.

All components (forward pass, backpropagation, and optimization) were implemented entirely from scratch in NumPy.

Dataset Overview: Synthetic Tasks



Gaussian Task — blobs well-separated; centroid gap $\gg$ within-class spread; linear boundary is LDA-optimal.



Moons Task — interlocking crescents; classes overlap in every axis-aligned projection; no affine hyperplane separates them.

Dataset Overview: Digits Benchmark

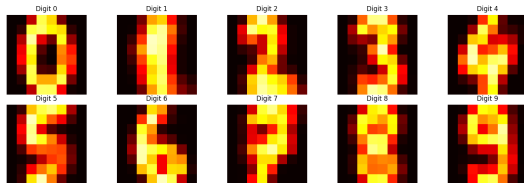
10 Classes (0–9), $d = 64$ features (flattened 8×8 pixel images)

Sample Images from Each Digit Class (train set, 5 examples each)



5 random examples per class from the train set.

Per-Class Mean Images (train set)
Bright pixels = high average intensity for that digit



Per-class mean images (bright = high average pixel intensity).

Split: 1,257 train / 180 val / 360 test (same shared indices for all teams).

Ensuring a Fair Comparison

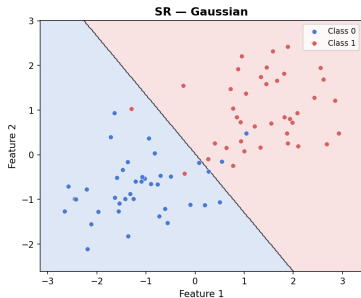
Because the models differ in structure and parameterization, our comparison must be mathematically fair.

Our Experimental Contract:

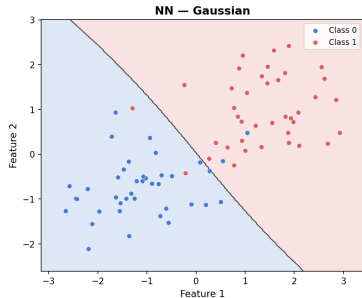
- **Identical Splits:** Fixed Train / Validation / Test indices across all runs.
- **Identical Preprocessing:** Digits used exactly as provided (scaled to $[0, 1]$), with no task-specific feature engineering.
- **Strict Model Selection:** Settings and checkpoints chosen *only* based on Validation Cross-Entropy.
- **Evaluation Metrics:** Unregularized Test Cross-Entropy (measuring predictive distribution quality) and Test Accuracy.
- **Statistical Rigor:** Final evaluations on the digits benchmark use 5 repeated random seeds.

When is Linear Structure Sufficient?

Task: Linear Gaussian (2 classes, 2D)



Softmax Regression

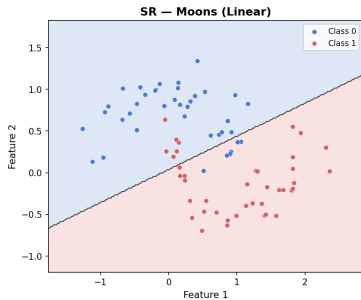


Neural Network ($m = 32$)

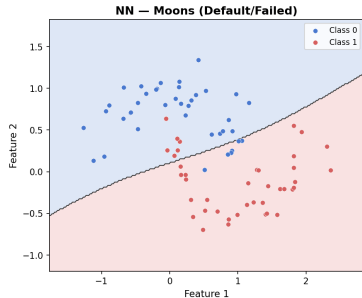
Conclusion: When the true Bayes-optimal separator is a hyperplane, adding a hidden layer provides **no meaningful advantage**.

Moons Task: The Failure of Default Optimization

Task: Moons (Default: 200 epochs, $\eta = 0.05$)



Softmax Regression

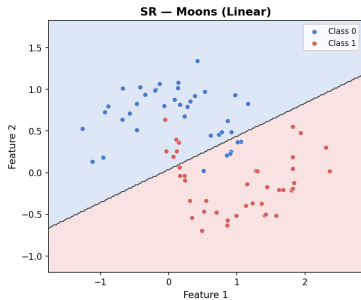


Neural Network (Default)

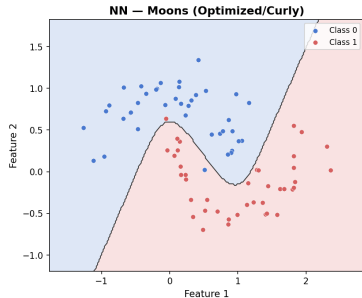
Observation: Both models produce nearly identical linear boundaries. Despite having 32 hidden units, the NN fails to learn the nonlinear structure because the learning rate and epoch budget are insufficient to escape the initial plateau.

Where Nonlinearity Helps

Task: Moons (2 classes, 2D interlocking crescents)



Softmax Regression



Neural Network (Changed/Optimized)

Conclusion: SR is **fundamentally limited** by its representation class — no choice of weights produces a linear boundary that correctly solves the moons task. The NN learns a curved separator by mapping the 2D input into a higher-dimensional space where the crescents become linearly separable.

Results on Digits Benchmark

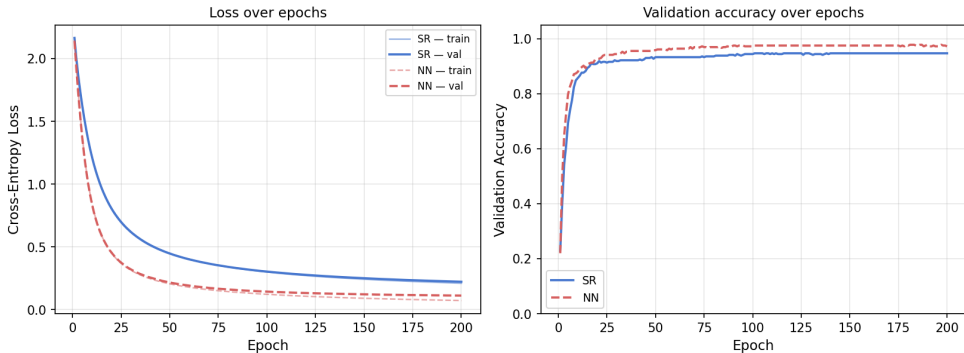
Task: Classify 8×8 pixel digits (64 dimensions, 10 classes).

Split (Seed 0)	Softmax Regression		1-Hidden-Layer NN	
	Accuracy	CE	Accuracy	CE
Train	0.965	0.202	0.991	0.063
Validation	0.946	0.214	0.972	0.103
Test	0.938	0.268	0.951	0.163

While Softmax Regression performs reasonably well, the Neural Network fits the training distribution more precisely and achieves superior generalization.

Digits Benchmark: Training Dynamics

Training Dynamics: Digits Benchmark



- The NN consistently achieves lower validation cross-entropy throughout training.
- Both models converge without severe overfitting within the 200-epoch budget.

What Repeated-Seed Evaluation Adds

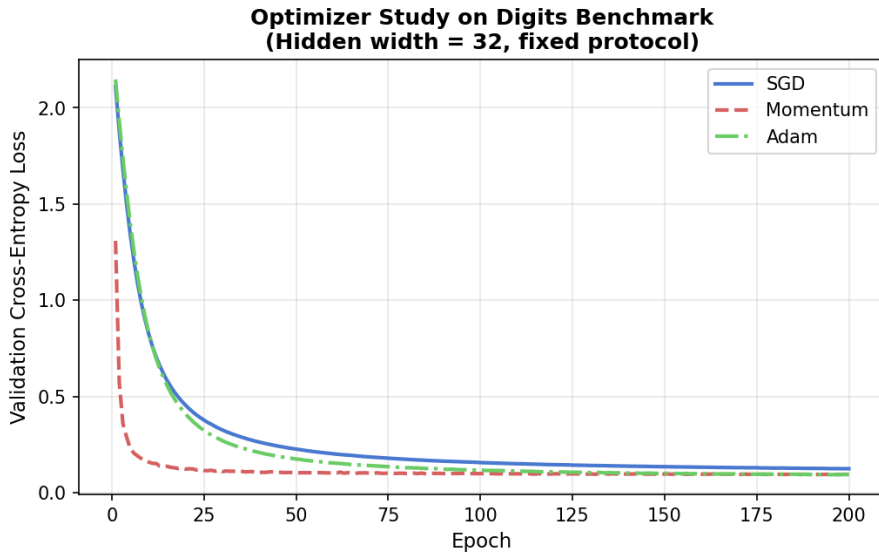
A single run can be lucky or unlucky due to random initialization. Re-running with 5 seeds provides statistical confidence.

Model	Test Accuracy (95% CI)	Test CE (95% CI)
Softmax Regression	0.9375 ± 0.0024	0.2683 ± 0.0027
1-Hidden-Layer NN	0.9511 ± 0.0000	0.1671 ± 0.0077

Takeaways:

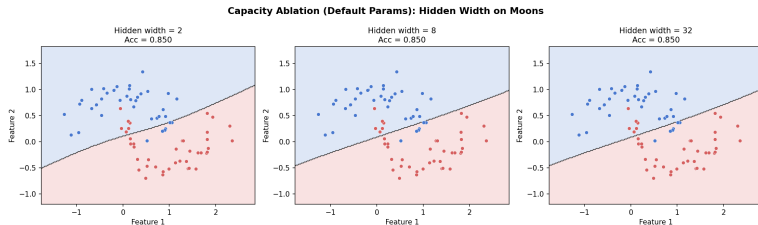
- Non-overlapping confidence intervals confirm the NN's advantage is statistically robust.
- SR is highly stable but bounded by its linear representational ceiling.
- The NN has higher variance in its predictive distributions (Cross-Entropy).

Optimizer Study on Digits (NN, $m = 32$, 200 epochs)



Failure Case Analysis: Optimization Limits

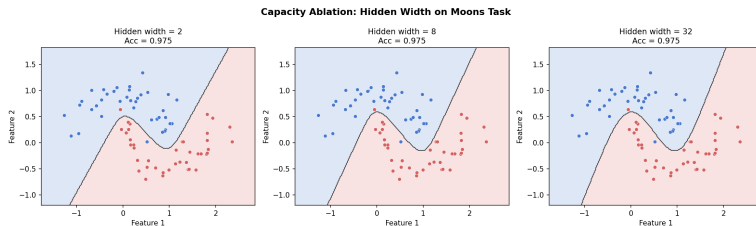
Experiment: Moons task, default constraints (200 epochs, $\eta = 0.05$).



- Near-linear boundary across *all* widths, even with $m = 32$.
- **Validation stall:** Accuracy stuck at 0.8875 for all widths ($m = 2, 8, 32$).
- **Why?** Default $\eta = 0.05$ is too slow to escape the loss plateau in 200 epochs.
- **Lesson:** Capacity without adequate optimization is useless.

Capacity Ablation: Complexity Is Not Always Better

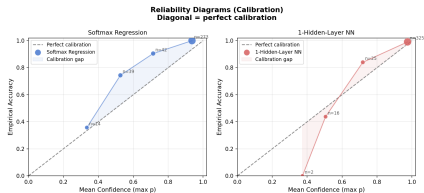
Experiment: Moons task, optimized training (1000 epochs, $\eta = 0.1$).



- Given proper optimization, validation precision reaches 0.9875 identically across all widths.
- **Key Insight: A highly complex model is not always better.** Introducing massive capacity ($m = 32$) provided no advantage over $m = 2$; the extremely lightweight network was fully sufficient.

Track B: Prediction Confidence and Reliability

Does the model know when it is likely to be wrong?



- **Softmax Regression** is systematically underconfident (empirical accuracy exceeds stated probability).
- **The Neural Network** is well-calibrated in its highest confidence bin ($Gap \approx -0.016$).
- **However**, although the NN has higher absolute accuracy, it can be more overconfident when giving wrong answers (e.g., 0% accuracy in the 20 – 40% bin).

Limitations of Our Study

- **Dataset Scale:** The digits dataset contains only 1,797 samples. Our findings may not seamlessly extrapolate to large-scale vision or language regimes.
- **Architectural Depth:** We analyzed exactly *one* hidden layer. We make no claims about the optimization dynamics or representational power of deeper networks.
- **Low-Dimensional Geometry:** The clear visual intuitions from the 2D Gaussian and Moons tasks do not perfectly mirror the geometric realities of high-dimensional feature spaces.

Summary

- **When is linear sufficient?** When the data geometry dictates an affine separator (e.g., Gaussian task).
- **What does the hidden layer add?** It warps the input space, allowing a linear output layer to solve nonlinearly separable problems (e.g., Moons, Digits).
- **Is complexity always justified?** No. If the mathematical optimization is inadequate (our failure case), or the task geometry is too simple, the extra complexity adds nothing.
- **Overall:** The One-Hidden-Layer NN outperformed Softmax Regression significantly on the digits benchmark in both raw accuracy and calibrated reliability.

Thank You!
Questions?